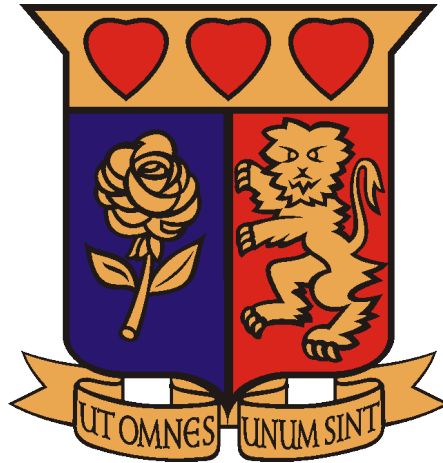




Strathmore University

Bachelor Of Science in Informatics and Computer Science (BICS)

Embedded Systems and Internet of Things (IOT)

ICS 4A

Practical Exercise 3

Group: Motion Masters

Group Members: Jimmy Kimotho, Daniel Tashibya, Alvin Omedi, Paul Obonyo, Sean Mudibo, Cyprian Kamau, Amy Zawadi (*left to right*)



Experiment 1

Objective: Practical demonstration of Ohm's Law.

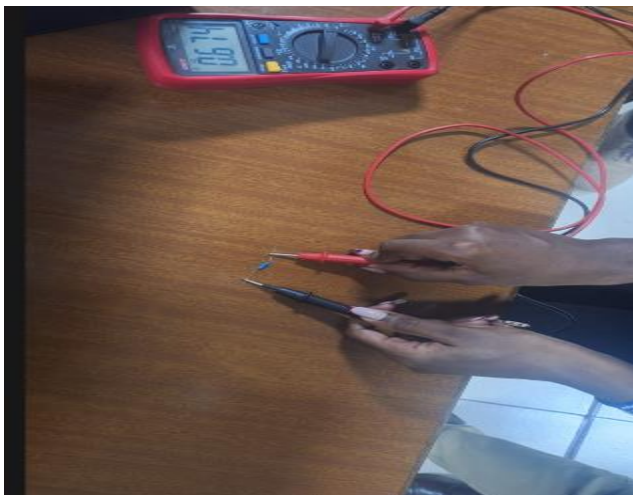
Task 1: We picked a resistor, took a picture of it and determined the resistance value based on the color code.



The resistor had the following color bands: Blue (6), Grey (8), and Brown ($\times 10^1$). The first 2 digits (6 & 8) were combined to give 68, and the last digit is the multiplier.

Therefore, the result will be **680 Ω**

Task 2: Using a multimeter, we measured the resistance of the resistor and noted its value



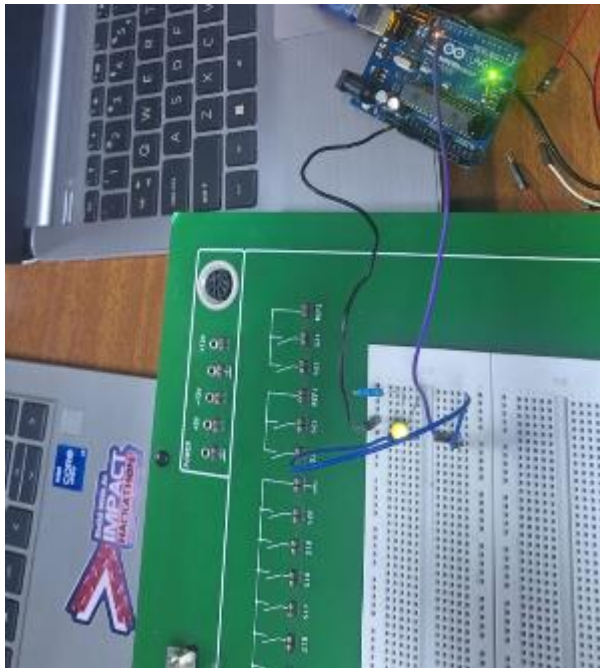
Measured resistance: **0.674 k Ω**

Task 3: Experiment Reflection: Where does the difference in the resistance values arise from?

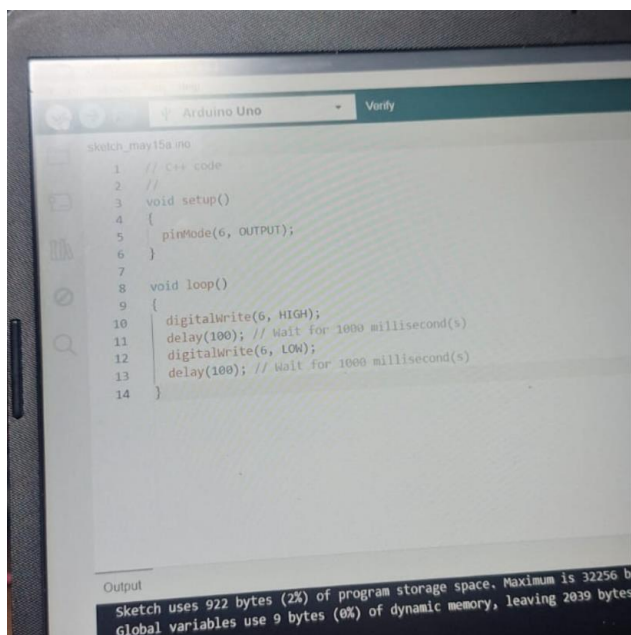
The difference arises because the color code gives the rated resistance, while the multimeter measures the actual resistance. Resistances have a tolerance level, meaning their true value can vary slightly from the rated value. Small manufacturing variations may affect the measured resistance, and therefore a slight difference between the two readings is expected.

Experiment 2:

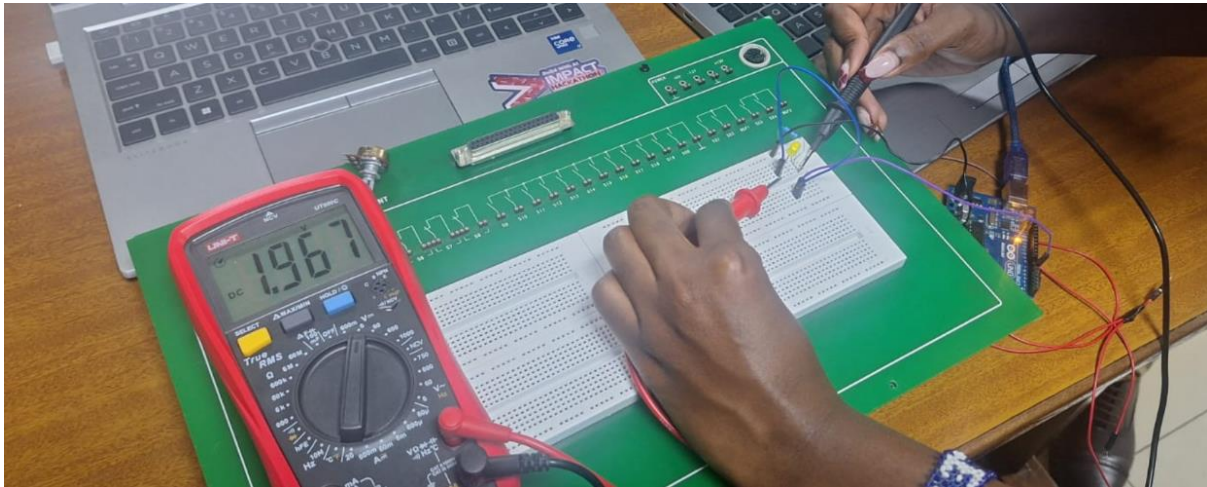
Task 1: Using a breadboard, microcontroller, and LED, we recreated the schema as per the design.



Task 2: We then uploaded the program to run on Arduino IDE

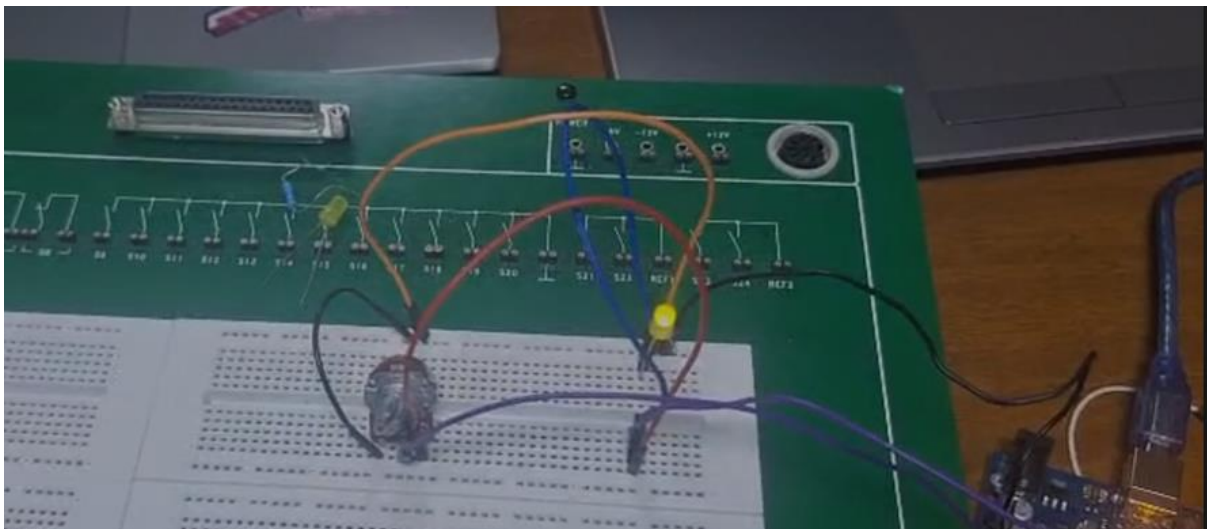


Task 3: We then used the Multimeter to determine the voltage passing through the circuit, obtaining a value of 1.967 Volts

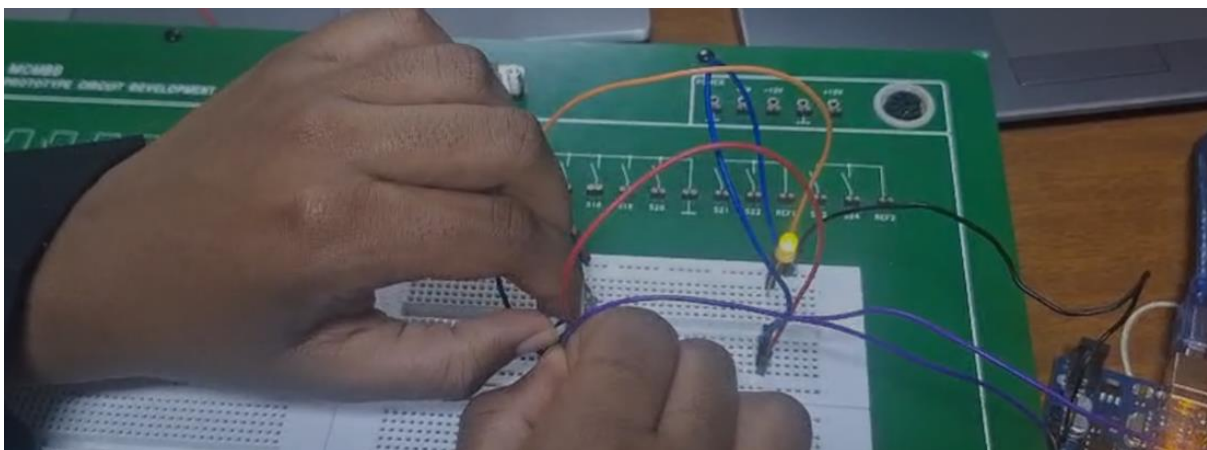


Task 4: We reconfigured the circuit by adding a potentiometer, as shown in the figure:

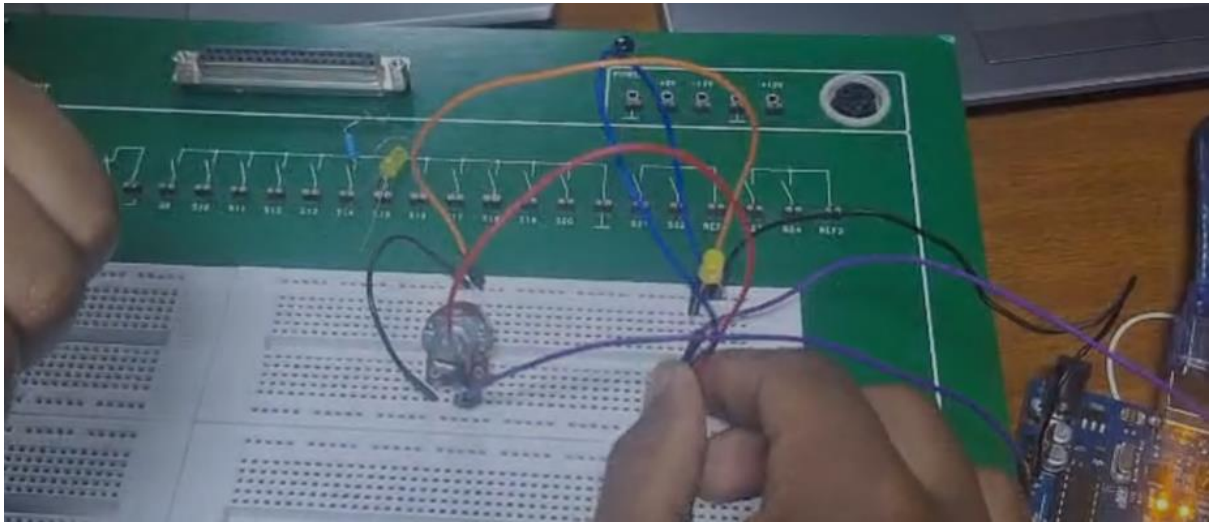
The image is at 100% brightness



The image is at 50% brightness:

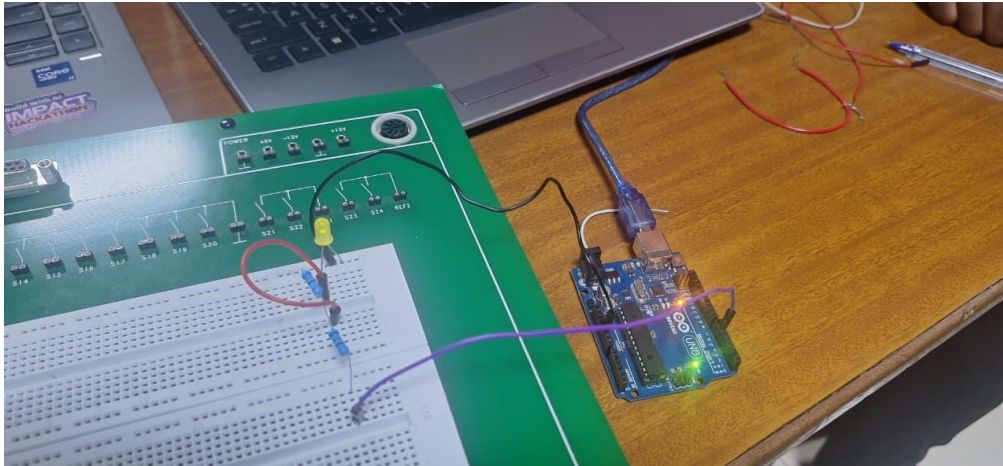


The image is at 0% brightness:

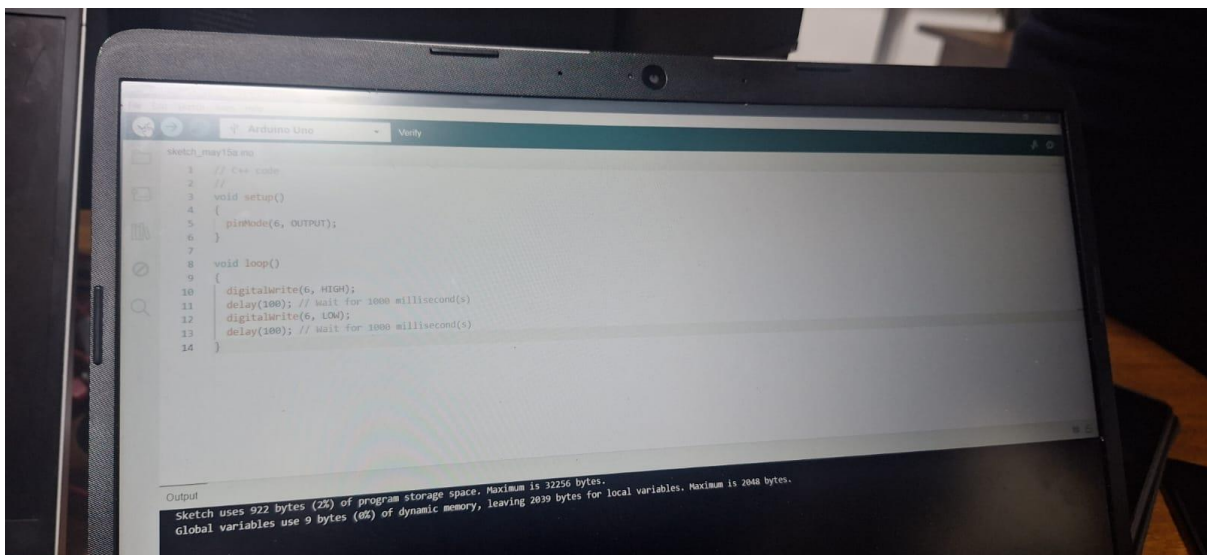


Experiment 3

Task 1: We assembled the circuit on a breadboard, according to the provided schematic



Task 2: We deployed the program in the Arduino IDE and uploaded to the Arduino board.



Task 3: We used a multimeter to measure voltage across each resistor in series





Task 4: finding the current passed at each resistor

$$R=680\ \Omega+680\ \Omega=1360\ \Omega$$

$$I=V/R$$

R1

$$I=2.538/680 = 0.00373\ \text{A}$$

R2

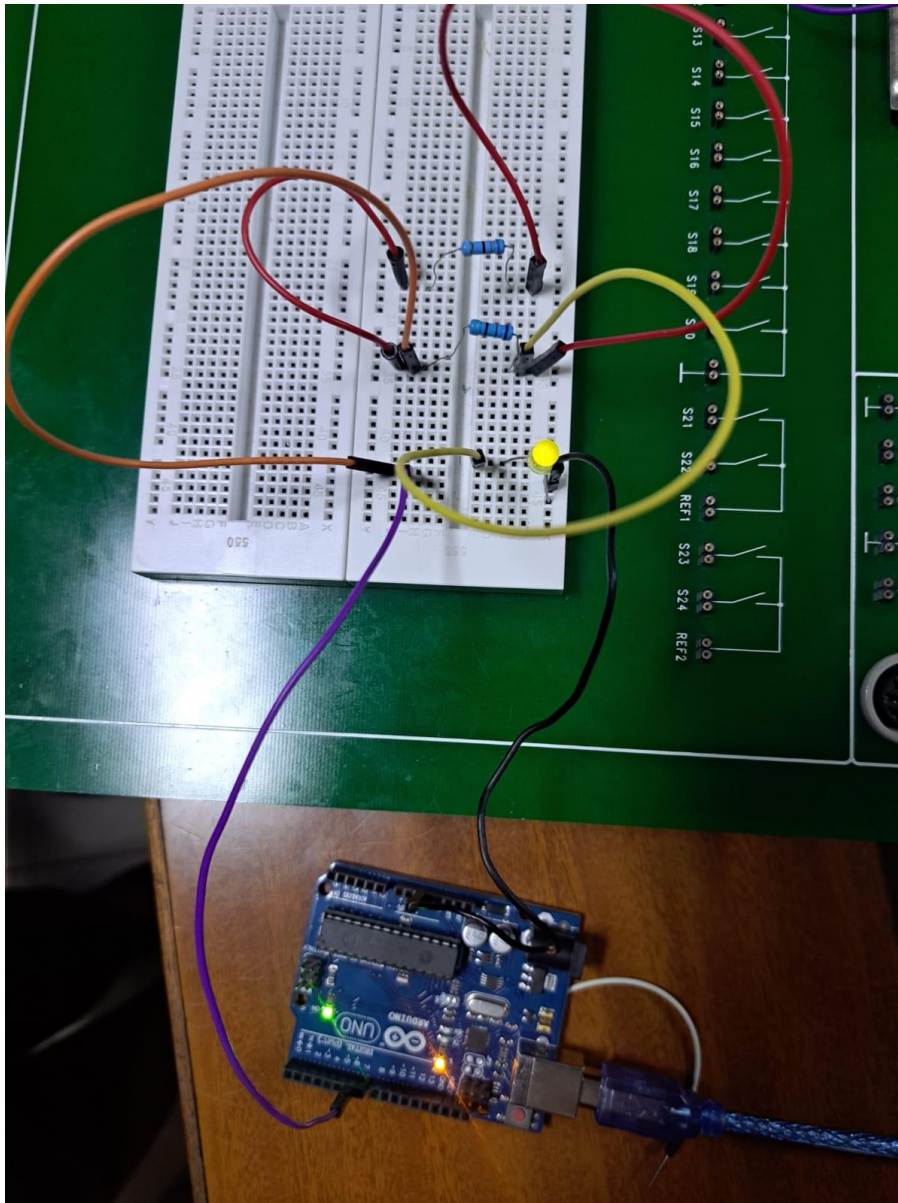
$$I=0.643/680 = 0.00095\ \text{A}$$

Experiment 4

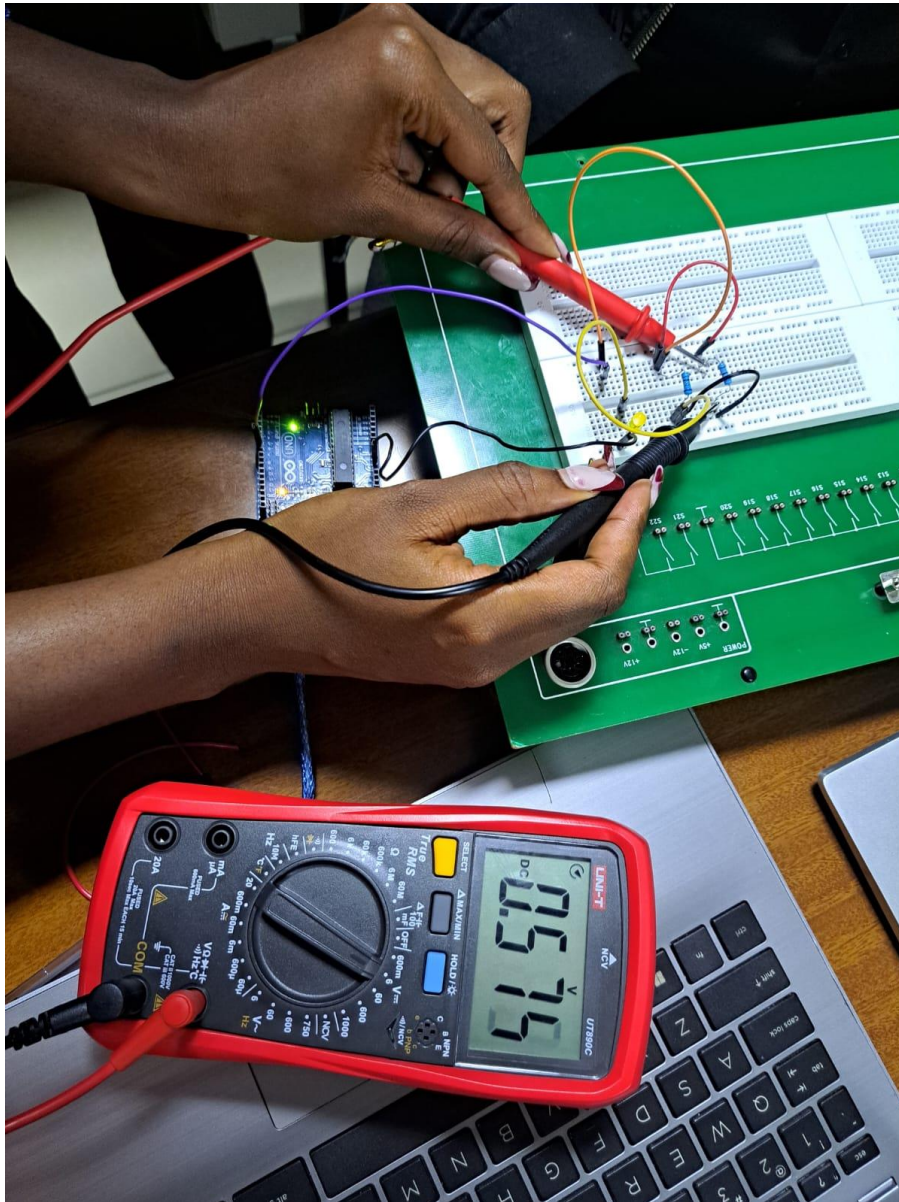
Objective: Use Ohm's Law to estimate the current passed at each resistor in parallel.

Task 1: With the Arduino microcontroller, wires, breadboard, resistors and an LED, we completed the circuit by connecting the resistors in parallel as per the schematic design as shown below.

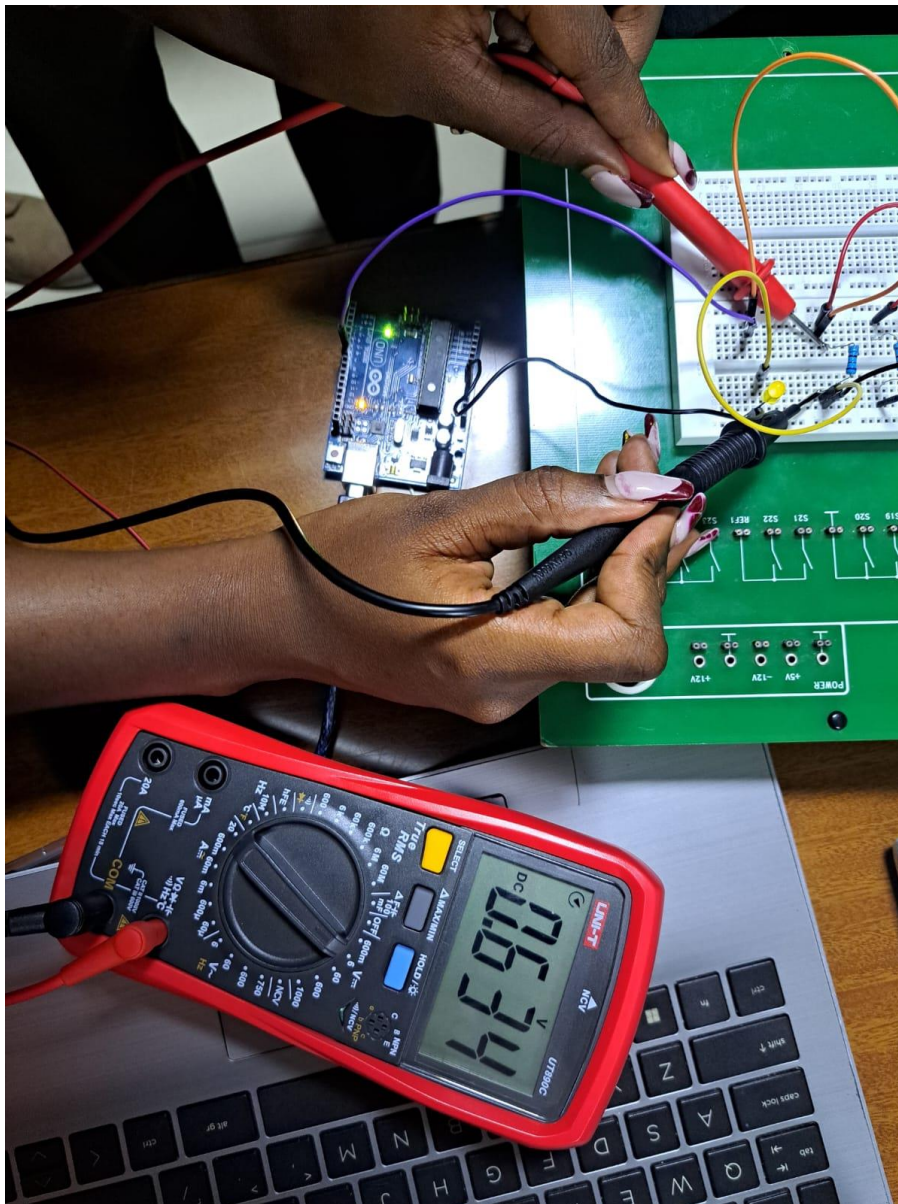
As a result, the Led bulb lit up signaling a successful circuit termination



Task 2: We measured the voltage passing through the furthest resistor which we recorded to be 0.57 volts



Task 3: Then we measure the voltage passing through the second resistor which was recorded to be 0.63 volts



Conclusion

With regards to Ohm's law, voltage is expected to be the same across all the resistors in parallel. However, based on our experiment we recorded a difference of about 0.06 volts which could be as a result of static charges detected by the multimeter, or probably a faulty multimeter was used for the experiment with a tolerance error of the same variance.

In terms of the observed voltage drop, this was likely due to the consumption of the LED connection in series and the circuit's internal resistance.

In summary, this experiment showed that the voltage remained the same when the resistors were connected in parallel, taking into account the standard margin error (0.06v).

